11th grade
Learning evidence 2.1: Confidence Interval Planning (Practice)

Evaluated criteria

This exam evaluates the following criteria. Each criterion is evaluated across the items below. A student passes a criterion if it is correctly applied in the solution.

- C1 Compute the sample mean and sample standard deviation, and summarize the sample data.
- C2 Distinguish between sample statistics and population parameters.
- C3 Explain why interval estimation is preferred over a single point estimate.
- C5 Construct a X% confidence interval using known population standard deviation.
- C4 Interpret the meaning of a X% confidence interval in context.

Problem 1: A distribution center monitors the time (in minutes) required to load outbound pallets. A sample of 75 loading times is grouped into class intervals below. The center has a historical estimate of the population standard deviation, $\sigma = 6{,}0$ minutes. For classroom reference, suppose the true population mean loading time is $\mu = 33$ minutes. The grouped intervals and frequencies are: 18–22 minutes (10), 23–27 minutes (16), 28–32 minutes (19), 33–37 minutes (17), and 38–42 minutes (13). Use the grouped data to estimate the sample mean and sample standard deviation. Construct and interpret a 95% confidence interval for the population mean loading time. Construct and interpret a 98% confidence interval for the population mean loading time.

Problem 2: A school cafeteria tracks the number of sandwiches sold during lunch to plan inventory. A random sample of 20 lunch periods is recorded below. The district provides a historical population standard deviation of $\sigma = 5{,}5$ sandwiches.

- Sample ($n = 20$): 42, 38, 45, 47, 40, 44, 39, 41, 46, 43, 37, 48, 40, 45, 42, 44, 39, 46, 41, 43.

Use the sample to compute the mean and sample standard deviation. Construct and interpret a 95% confidence interval for the population mean sandwiches sold. Construct and interpret a 98% confidence interval for the population mean sandwiches sold.